

Alert Management Practice

1. Alert Classification System

Alert ID	Alert Type	Description	Priority	MITRE ATT&CK Technique	Status
001	Phishing	Suspicious email with malicious link	High	T1566	Open
002	Brute Force	Multiple SSH login failures	Medium	T1110	Open
003	Malware	Trojan detected on workstation	High	T1204	Investigating
004	Ransomware	File encryption activity detected	Critical	T1486	Escalated
005	Port Scan	Multiple ports scanned from external IP	Low	T1046	Monitoring
006	Credential Dumping	LSASS memory access detected	Critical	T1003	Escalated
007	PowerShell Abuse	Encoded PowerShell execution	High	T1059.001	Investigating
008	Data Exfiltration	Large outbound traffic detected	Critical	T1041	Escalated

2. Alert Prioritization Worksheet

Scenario 1

Alert

Log4Shell Exploit Attempt Detected

Asset: Production Web Server

CVSS Score: 10.0

Business Impact: Very High

Exploit Available: Yes

Priority: **Critical**

Reason:

The vulnerability allows remote code execution and has publicly available exploits. The affected asset is business-critical.

Scenario 2

Alert

Port Scan Detected

Asset: Test Machine

CVSS Equivalent Risk: Low

Business Impact: Low

Priority: **Low**

Reason:

No active compromise detected and target is a non-production system.

Scenario 3

Alert

Multiple Failed VPN Logins

Asset: Employee VPN Portal

Priority: **Medium**

Reason:

Potential brute-force attempt requiring investigation.

3. Incident Response Procedure

Critical Alert Workflow

Step 1: Alert Detection

Alert generated by SIEM (Wazuh).

Example:

Ransomware activity detected

Step 2: Initial Verification

Analyst verifies:

- Source IP
- Destination IP
- User account
- IOC validity

Step 3: Containment

Actions:

- Isolate infected host
- Disable compromised account
- Block malicious IP

Step 4: Escalation

Notify:

- Tier 2 Analyst
- Incident Response Team
- SOC Manager

Step 5: Investigation

Collect:

- Logs
- Memory dump
- Network traffic
- File hashes

Step 6: Remediation

- Remove malware
- Patch systems
- Reset credentials

Step 7: Recovery

- Restore backups
- Verify functionality
- Monitor systems

Step 8: Lessons Learned

Document:

- Root Cause
 - Timeline
 - Impact
 - Recommendations
-

4. Sample Incident Ticket

Incident ID

INC-2026-001

Title

[Critical] Ransomware Detected on Server-X

Date & Time

29 May 2026 – 11:00 AM

Description

Wazuh detected ransomware activity on Server-X. Several files were encrypted and unusual process execution was observed.

Indicators of Compromise (IOCs)

Malicious File:

crypto_locker.exe

Attacker IP:

192.168.1.50

Hash:

SHA256:

8c6976e5b5410415bde908bd4dee15df

Affected Asset

Server-X

IP:

192.168.1.10

Priority

Critical

Status

Escalated

Assigned To

SOC Tier-2 Analyst

Recommended Actions

1. Isolate server.
 2. Collect forensic evidence.
 3. Remove malware.
 4. Restore affected files.
 5. Conduct root cause analysis.
-

5. Escalation Email

Subject:

URGENT: Critical Security Incident Escalation – Ransomware Detected

Dear Tier-2 Response Team,

A critical security alert has been identified involving ransomware activity on Server-X. Initial investigation confirms suspicious file encryption and malicious process execution.

Key Indicators:

- Affected Server: Server-X
- Malicious File: crypto_locker.exe
- Source IP: 192.168.1.50
- Alert Severity: Critical

Immediate containment measures have been initiated, including host isolation. Further forensic investigation and remediation are required.

Please prioritize this incident and provide updates regarding containment and recovery actions.

Regards,

SOC Analyst

Cybersecurity Operations Team

6. Alert Escalation Matrix

Priority	Escalate To	Response Time
Critical	SOC Manager + Incident Response Team	Immediate
High	Tier 2 Analyst	Within 1 Hour
Medium	SOC Analyst	Within 4 Hours
Low	Monitoring Queue	Within 24 Hours

7. Alert Dashboard Summary (for Screenshot/Report)

Priority Count

Critical 4

High 8

Medium 12

Low 20

Total Alerts: **44**

Observations

- Critical alerts require immediate attention.
- Most alerts are low-priority informational events.
- Phishing and malware incidents account for the majority of high-priority alerts.